

Object recognition in refrigerator scenario

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Problem & Goal

Problem Statement:

General-purpose object detection models perform poorly in refrigerator environments due to cluttered layouts, limited space, lighting variability, and visually similar packaging. These challenges hinder accurate identification of stored food items, essential for smart kitchen systems.



Training Image
Ref from: Roboflow

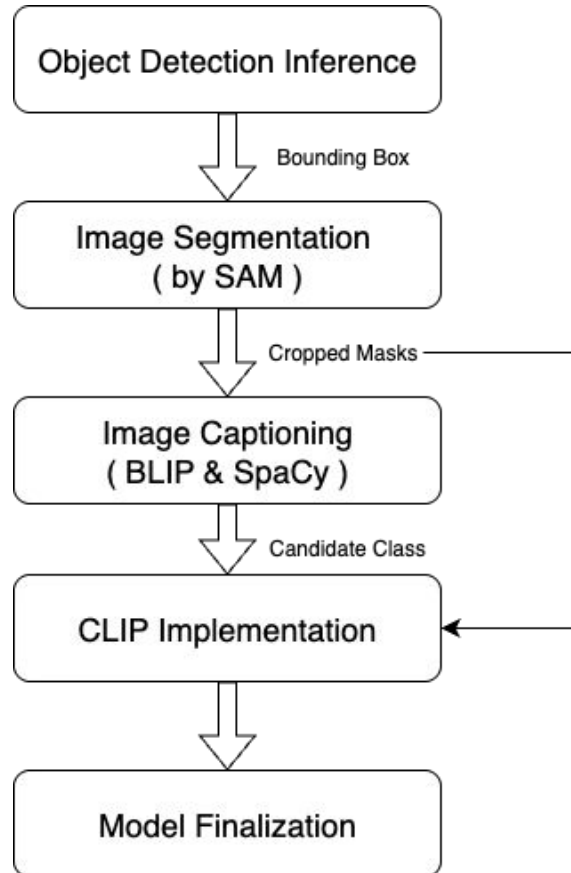


Test Image

Goals:

1. Develop specialized object recognition models tailored for refrigerator environments.
2. Improve detection under occlusions, constrained viewpoints, and poor lighting.
3. Attempt to enable accurate food item classification to support automation, reduce waste, and promote health-aware lifestyles.
4. Push the task to open-world task to detect unknown categories in dataset.

Overview of approach



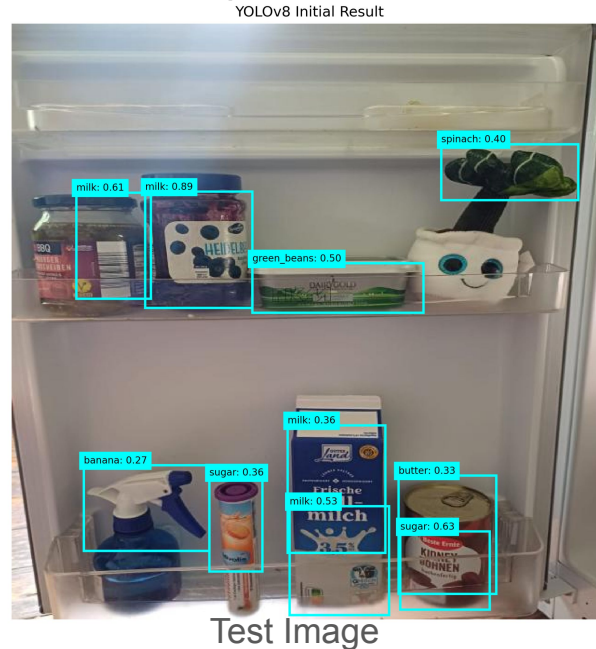
Related Work

1. [Arxiv \(IEEE Xplore\). Aerial Image Segmentation Using YOLO v8 and SAM Pipeline, 2023](#)
2. [Food detection system in the refrigerator based on YOLOv10 deep learning yolo food dataset + ui interface + model + real-time detection + 30 kinds of food](#)

Detailed Approach

Step 1: Object Detection Inference

Use pre-trained YOLOv8 on a refrigerator dataset containing 3000 images and 30 known food classes to generate rough bounding boxes and initial predictions



Step 2. Image Segmentation by SAM

Pass bounding boxes to SAM to extract more precise cropped masks corresponding to each object.

YOLOv8 Initial Result



Step 3: Image Captioning

1. BLIP to get descriptions of masks.
2. NLP tools (SpaCy) to extract all potential adj+noun./noun. and get rid of common words.
3. Finally generate a list of potential class.



BLIP Description: a green leaf on a black background

NLP Extracted Candidates: ['blackbackground', 'leaf', 'background', 'greenleaf']

["item", "object", "thing", "piece", "top", "front", "back", "side", "some", "a", "an", "the", "bunch", "pile"]

Both images are from the previous test image

Step 4: CLIP Implementation

Pass cropped masks from SAM and candidate class from BLIP to CLIP and get predictions with highest probability

Step 5: Model Finalization

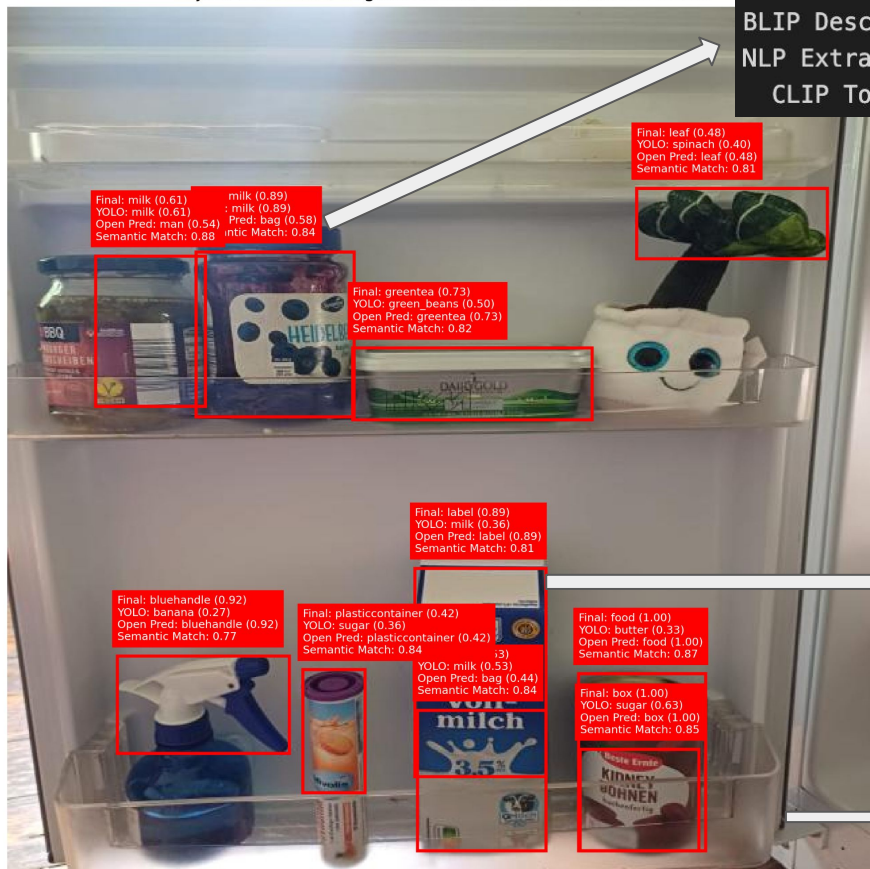
Compute the semantic similarity(cosine similarity) and compare the confidence of YOLO and our model.

If the confidence of YOLO is higher and semantic similarity is higher than threshold we accept YOLO prediction. Otherwise, we prefer our model.

```
YOLO_HIGH_CONF_THRESHOLD = 0.70  
SEMANTIC_MATCH_THRESHOLD = 0.75
```

Results

Object Detection, Segmentation, and Fused Classification



BLIP Description: a bag of blueberries on a black background

NLP Extracted Candidates: ['bag', 'blueberry', 'background', 'blackbackground']

CLIP Top Prediction (open-ended): bag (0.5767)

BLIP Description: a box of milk with the label on it

NLP Extracted Candidates: ['milk', 'box', 'label']

CLIP Top Prediction (open-ended): label (0.8923)

BLIP Description: a box of king queen chocolate

NLP Extracted Candidates: ['box']

Conclusion

Successfully predict classes not included in known class list: leaf, blue handle, plastic container.

Drawbacks: Our model prefer to label objects as containers instead of contents (limitation of CLIP and BLIP as we have not fine-tuned it)

Scope of Improvement: We are planning to fine-tune CLIP and BLIP for better class detection.

Thanks for your attention!